



Decisional architectures from business intelligence to Big Data: challenges and opportunities

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Abstract

Information is one of the most important factors in business success, hence the importance of the Business Intelligence (BI) domain in order to simplify the decision making and make it more relevant. Decisional systems have been used for several years to help decision-makers access, analyze and extract value from the data that their organisation accumulated through the years. The success gained by these types of systems caused the establishment of a well-known architecture and development chain, and the proliferation of tools and methodologies that have proven their value. Nonetheless, in some use cases, the classical decisional architecture shows some shortcomings. In fact, the traditional storage and processing models in Business Intelligence systems are not sufficient anymore when confronted with data that becomes more and more massive, varied and with a high velocity. This is where Big Data solutions can be of great use. In fact, these solutions have proven their efficiency when dealing with enormous constantly increasing amounts of data with a changing schema. Our goal in this article is to show the various manners to integrate big data solutions into the decisional world, and to help architects choose which architecture corresponds better to their needs, by taking into consideration the environmental, technical and functional constraints they are faced with.

CCS Concepts: • **Information systems** → **Data warehouses; Mediators and data integration; Distributed storage; Data analytics; Enterprise applications;** • **Software and its engineering** → **Software architectures.**

Keywords: Business Intelligence, Big Data, Decisional Systems, OLAP, Software Architecture, Decision Tree for Decisional Architectures, Data Lake, Data Warehouse

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1 Introduction

Nowadays, information represents the main source of competitiveness and income for business success. For this reason, companies wish to hold the most effective tools for data analytics and management in order to make the best decision at the right time, in a market where the competition is increasingly stiff and where having the most reliable information means you are ahead. Thus, organizations must act with regard to the exponential increase of data volume. Moreover, they must ensure the good governance of said data by analyzing it rapidly and efficiently, with the aim of exploring its environment and its evolution in order to acquire a unique competitive advantage.

The problem with data is that acquiring it is quite easy, as the amount of relevant and irrelevant data exchanged, stored and generated by the modern technologies is increasing in an astronomical way, but extracting some *value* from it happens to be a very tedious operation. In fact, the main concern of data engineers has shifted from how and where to store data, to how to process it in order to transform it to valuable information, right on time, and with as little data loss as possible. In order to do this, a panoply of technologies, paradigms, tools and platforms are invented everyday, and system architects and developers are completely lost between them. For this reason, architects need a reference guiding them on the existing architectures, the alternatives and some choice-making pointers on which solution to adopt. Unfortunately, documentation is lacking in this field, and this is the purpose of our current work.

Our main goal in this paper is to gather information that encourage engineers, computer scientists, data architects and researchers by presenting, classifying, explaining and comparing the existing possible architectural patterns a software architect can adopt in order to set up a decisional system, depending on a set of constraints.

We present these patterns and explain their respective benefits, limits and some possible examples of use. We propose then a decision process that anyone can follow in order to help him/her make a choice to find the adequate architecture for his/her needs, depending on the data and the processing types.

2 Big Data and Business Intelligence

Business Intelligence applications use tools, techniques and technologies allowing the exploitation and the consolidation of the heterogeneous data, in order to facilitate decision making within the company and to allow the decision maker to have an overview of his activity. They based on Online Analytical Processing (OLAP) that structures data as a set of facts and dimensions. They rely on a well defined process: data extraction, transformation, loading into the data store, analysis and visualisation. These steps will be further detailed later in the article.

On the other hand, Big Data technologies allow collecting, storing and analyzing massive data characterized by volume, variety and velocity via varied terminals, in real time and from any geographical zone. They represent, today, powerful tools for companies in the decisional world thanks to their distribution, scalability and modern architectures. The introduction of these technologies gives companies many opportunities to improve their information system and to make it more efficient by supporting new features and options that did not exist before. Big Data systems use distributed storage systems to benefit from their (theoretically) unlimited capacity, and the computation power they can offer thanks to the multiplication of resources. Technologies differ in this area, from those that enable the processing of huge quantities of inputs in order to have a complete view of the existing data and extract value with a high throughput, to those that process arriving high velocity data in real-time to render results at a low latency. Big Data technologies can be integrated in every layer of the information system, and in all types of applications: transactional, analytical, or transverse, such as monitoring, business process management, and security. In this article, we are only dealing with OLAP architectures and the impact of Big Data technologies on them, which introduces a new OLAP concept known by **Big-OLAP**. In fact, Big Data systems can also be used in the decisional world, as they provide powerful and precise tools for extracting value and facilitating decision-making and prevision. Some people even tend to declare that, now that Big Data is evolving and maturing, the classical Business Intelligence is about to disappear. The questions now being asked are whether this assumption is accurate. Are classical BI systems doomed to become extinct? Can Big Data solutions be the response to all decisional needs?

Contrary to BI solutions, Big Data systems don't follow a rigid and well-defined decisional chain, but are characterized

by the flexibility and adaptability of their architectures. Moreover, Business Intelligence applications are always driven by KPIs (Key Performance Indicators), contrary to Big Data solutions that support a schema-less approach applicable for all decisional KPIs and requests[9]. Decisional Big Data systems are known to be data-driven, which means that the business needs are extracted from the existing data, and can evolve with them. While BI systems work the other way round: the business needs (illustrated by KPIs) are defined first, and from these needs, the structure and type of the stored data are deduced. On the other hand, storing data into a BI system must go through a rather heavy and long transformation phase that must happen periodically, preferably at night or in the weekend when the source systems are at rest. But in Big Data systems, and due to the velocity property of the data to be stored, subjecting them to a long processing step before storage is quite impossible. This is why they need to be stored first, as they are. This important distinction drives the need to very different types of storage systems: for Business Intelligence applications, we are talking about Data-Warehouses, versus Data-Lakes for Big Data systems.

The Data-Warehouse is a structured store that respects either a star, snowflake or fact-constellation schema. It is a centralized, structured and KPI-driven storage system. On the other hand, a Data-Lake is a data store designed for Big Data applications, characterized by its distribution and capacity to store and treat a high volume of data. In addition, the most important advantage of a Data-Lake is its capacity to manage a schema-less data that is often denormalized, redundant and stored in bulk. The choice of the terminologies "Data-Warehouse" and "Data-Lake" explains very clearly the main difference between these two notions. Being a warehouse, a Data Warehouse is there to store the data in an organized manner meant to facilitate searching and extracting valuable information. Whereas a lake is there just to engulf all the goods you intend to throw in it. It doesn't care about classifying it, filtering it or structuring it at first. It is just intended to store a huge amount of data, and it's up to the client later to go through all the mess and see what he can do with it. In Data-Warehouses, data needs to be shaped before it is stored, as it uses a *schema-on-write* processing type, while Data-Lakes do not use any predefined structure while writing data, but defines the necessary structure for processing when reading it (*schema-on-read*). In Data-Warehouses, data is generally read using multi-dimensional queries and transformed into in-memory hyper-cubes before being sent to the reporting tool, while with Data-Lakes, treatment is co-localized with the data, supports distribution and parallelization, and can be pretty complex.

Both data stores are historical, which means that data is not supposed to be deleted or modified, only added and read. All data are tagged along with timestamps, and any modification or new addition will just be appended to the old

value, thus enabling archiving and applying a next-to-zero-data-loss strategy. As a summary, we present in the Table 1 a more exhaustive comparison between these storage systems.

3 Decisional architectures from Business Intelligence to Big Data

In this work, our first goal is to list and distinguish between the different decisional architectures starting from pure BI down to pure Big Data solutions.

3.1 BI Architecture

The Business Intelligence Architecture is the traditional and most used decisional architecture, characterized by its four famous phases: The first one is Extraction, Transformation, Loading (ETL) and in this phase, data is extracted from different sources, transformed and loaded into the Data-Warehouse. ETL jobs are planned with low frequency and has generally a high input data volume. The second phase is the Data-Warehousing and it consists in storing data, modeled according to the appropriate KPIs defined by the decision makers, into different fact and dimension tables designed either using the star schema, snowflake schema or fact constellation schema. The next phase is the data analysis and in this phase the Data-Warehouse is queried with advanced SQL or multidimensional queries, thus creating in-memory OLAP cubes, in order to extract the designed KPIs. The final phase is the data visualization and it consists in representing analysis results with graphs, tables, reports or dashboards.

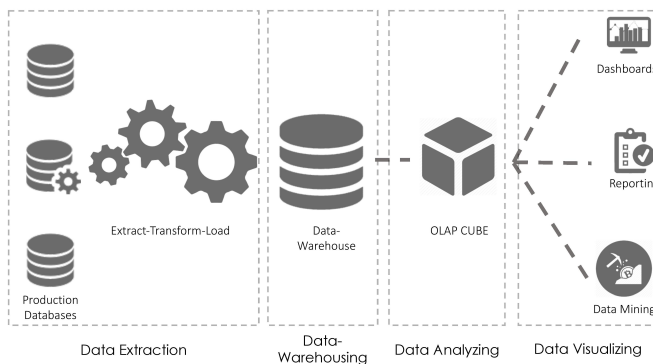


Figure 1. Business Intelligence Architecture

Benefits: One of the benefits of the Business Intelligence architecture is its simplicity, its straightforwardness, its facility of design and implementation, and the fact that it presents

¹The information density describes the proportion of the useful data in the whole data set.

²Vertical scalability means that you scale your system by adding more power (CPU, RAM) to an existing machine, while horizontal scalability means that you scale it by adding new machines to the pool of resources.

the organization's information consistently. It is also a rather mature architecture that has already proven its worthiness, and has benefited for several years from the undivided interest of the decisional community, which provoked the rise of several methodologies, tools and good practices, widely used in industry these days.

Limits: The Business Intelligence Architecture has already shown its benefits in a lot of use cases, but has some limitations when it comes to dealing with data that can be categorized as "Big Data". Let's first consider the first particularity of Big Data: volume. It is true that a Data-Warehouse is destined to be a warehouse for a huge amount of historical data, but it doesn't scale well to data which volume increases steadily and continuously. This is due to the fact that only vertical scalability is supported, which makes its maximum capacity quite limited, and calls for very expensive hardware. As to the "velocity" characteristic, the BI architecture doesn't support real time processing, which is very important if we want to process or visualize data on the fly. It also does not support distributed processing and can only run planned jobs that cannot support streaming data sourcing. The "variety" property means not only various data sources, which is supported by this architecture, but also changing and evolving data types, which is less so. In fact, the Data-Warehouse is designed according to certain defined KPIs. But while these KPIs can change frequently, it is difficult to redesign the Data-Warehouse in the case of a new decisional need. We can also highlight other drawbacks, such as the loss of a part of data after the ETL phase, or the sole existence of static and non-interactive dashboards.

Compatible technologies: According to the "2017 Magic Quadrant for Business Intelligence and Analytics Platforms" of Gartner [13], the leaders and most used BI and analytics platforms are QlikView, Microsoft BI and Tableau. We can also cite some open-source suites, such as Jaspersoft or Pentaho [10].

3.2 Big-ETL

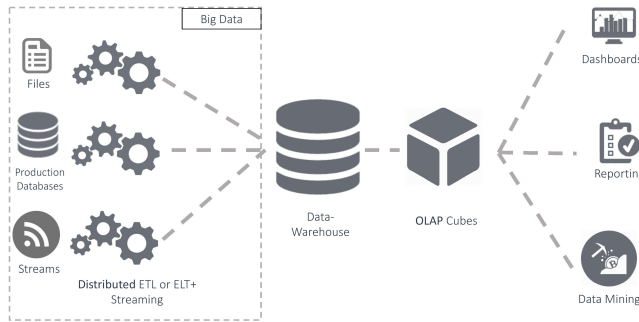
In this architecture, we observe the first appearance of Big Data technologies in the decisional architecture. Simple planned ETL jobs are replaced by jobs using Big Data ETL tools which are characterized by a distributed architecture based on streaming or on planned jobs.

Benefits: Being distributed, the extraction phase becomes faster and can be triggered more frequently, avoiding the long ETL batch phase, which is the main cause of the version gap between the data sources and the Data-Warehouse. This solution is also characterized by the support of data streams collection and real-time transformation.

Limits: The Big-ETL architecture can only remedy the problem of periodic and heavy ETL processing, but can cause a limitation in the loading phase. In fact, as the Data-Warehouse is still centralized, it will be subject to a high congestion rate coming from several parallel streams from

Table 1. Data-Warehouse VS Data-Lake

	Data-Warehouse	Data-Lake
Data Structuring	Structured	Not (necessarily) structured (denormalized)
Subset	Data-Mart	Lake-Shore
Data Volume	Important	Very High ($\geq 100TB$)
Information Density ¹	High	Low
Distribution	Centralized	Distributed
Co-localized Treatment	Not supported	Supported
Scalability ²	Vertical	Horizontal and Vertical
Users	Business Professionals and Decision Makers	Data Scientists and Data Engineers

**Figure 2.** Big-ETL Architecture

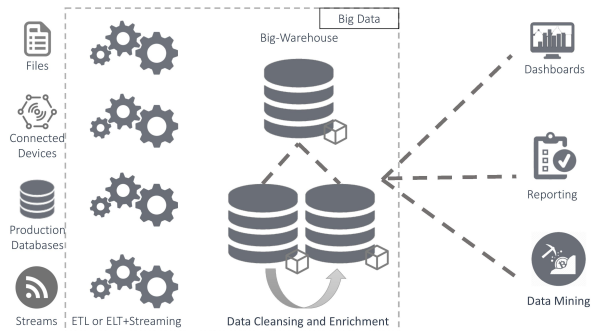
various data sources. On the other hand, one of the advantages of using the distributed ETL would be to enable one-to-one ETL processes: each process will perform a transformation from one data source to a unique physical part of the Data-Warehouse, thus enabling real and effective parallelism. However, this very handy approach can only be possible if the Data-Warehouse was physically distributed, which is not the case in this architecture. On the other hand, applied as it is, homogenizing the data coming from various data sources will become harder, due to the difficulty to manage shared states between distributed processes.

Compatible technologies: Traditional ETL tools are beginning gradually to integrate the era of Big Data technologies. Among them, we find Pentaho [12], which supports Big Data architectures and technologies by integrating new plugins and features that simplify distributed data integration from/to traditional or Big Data platforms[14]. On the other side, Talend created a new product named Talend Big Data Integration that replaced traditional ETL components with Big Data engines like Spark and MapReduce[16].

3.3 Big-Warehouse

One of the big problems of the Data-Warehouse in the classical decisional architecture, is that it doesn't support the Big

Data volume, that can impact the complexity of data restoration when dealing with a large number of joins. In fact, the more the data volume increases, the worse the read performance is. For example, if we consider two tables A (of size M) and B (of size N), the join complexity can even reach $O(MN)$, in the case of the absence of indexes, which is quite frequent in large scale databases. Thus, in order to resolve this kind of problems, the Big-Warehouse architecture uses Big Data technologies in the warehousing phase. Big-Warehouse is characterized by a distributed architecture that can support both processing types: ETL and ELT (Extract-Load and Transform). Similarly to traditional Data-Warehouses, Big-Warehouse preserves metrics and dimensions concepts but with a different data model based on data de-normalization which reduces join operations.

**Figure 3.** Big-Warehouse Architecture

Benefits: The most important benefit of the Big-Warehouse architecture is that it is vertically and horizontally scalable, which will enhance its ability to store and process a high volume of data with a reduced cost. It can easily be integrated with analytics and ETL Big Data tools, thanks to their adequacy with distributed architectures.

Limits: Despite the benefits of the Big-Warehouse architecture, it doesn't support real time stream processing and historical analytics because it resolves only the high data

volume problem of the traditional Business Intelligence architecture, without dealing with its velocity and variety, as our warehouse is still very structured and our processing still KPI-oriented and planned.

Compatible technologies: This architecture is based on a de-normalized and distributed data storage and (1) a real time, interactive and distributed SQL data processing (such as Apache Impala [5], Apache Presto and Apache Drill[4]), or (2) a Big Data Data-Warehouse infrastructure with orchestrated jobs (such as Apache Hive [17]), both plugged to BI tools. One of the most used technologies that respect the Big-Warehouse architecture is Apache Kylin. Indeed, it is an open source distributed analytics engine and Big Data storage designed to provide multi-dimensional analysis on Hadoop, supporting extremely large datasets using Hive and HBase [7].

3.4 Data-Lake $\cup$ Data-Warehouse

This approach consists of coupling both a Data-Warehouse and a Data-Lake in the same architecture, instead of choosing one of them, benefiting thus from both structures. We identified two main ways to integrate a Data-Lake with a Data-Warehouse:

Data-Lake in series with the Data-Warehouse (Sequential-Architecture: Fig 4). In this case, the Data-Lake represents a sort of bridge between the production data sources and the Data-Warehouse. The data stored in the Data-Lake is not necessarily cleansed or filtered and it is characterized by a high volume and a low *density*. An ETL process is then implemented to periodically transport data from the Data-Lake to the Data-Warehouse to be analyzed and visualized by BI tools. In this part of the process, data is characterized by a (relatively) low volume, high density and rigid structure.

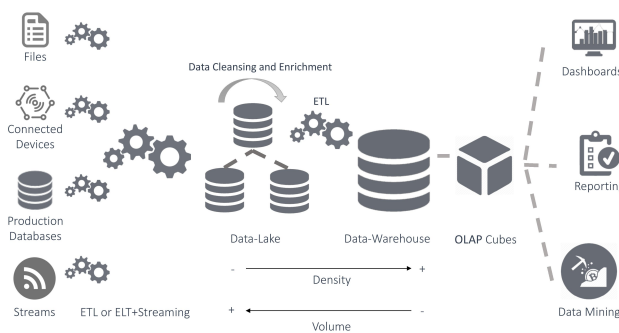


Figure 4. Data-Lake $\cup$ Data-Warehouse - Sequential Architecture

Data-Lake in parallel with the Data-Warehouse (Y-Architecture: Fig 5). In this architecture, the Data-Lake is placed in parallel with the Data-Warehouse to store the same or a different scope of data. For example, for a trading

application, the data lake can be used to store social media streams, while the Data-Warehouse will store only financial and trading data. When scopes match, the Data-Lake will store the totality of data for historic analysis and the only similarity in the ETL process is in the extraction phase. Otherwise, stored data is not the same and the Data-Lake will be used solely for Big Data use cases.

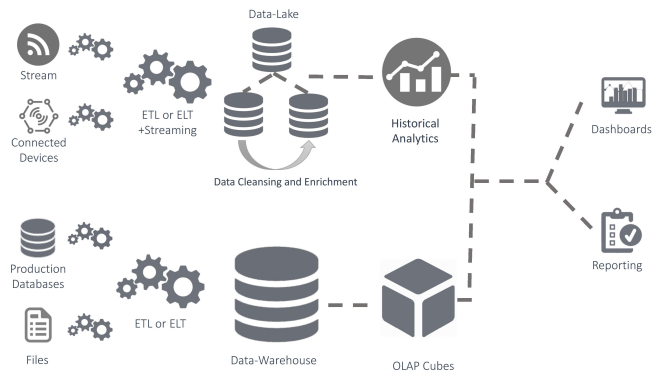


Figure 5. Data-Lake $\cup$ Data-Warehouse - Parallel Architecture

Benefits: In this architecture, we combine the advantages of both domains: there's no data loss caused by the ETL process, therefore, it supports historic analysis thanks to the Data-Lake. Also, it supports streaming data sources and ELT processing and it can be easily integrated with the decisional legacy systems.

Limits: Among its limits, this architecture can lead to data incoherence between the Data-Lake and the Data-Warehouse. A stricter data governance is necessary to avoid such inconsistencies, and still can be harmful with regard to performance and availability. Joins between the Data-Lake and the Data-Warehouse are not efficient because of the difference in performance between these systems. The Data-Lake is fully distributed and scalable, which increases its capacity to execute read and write operations with low latency, contrary to the Data-Warehouse, which is not. These architectures, even when they resolve scalability, data-loss and bottleneck problems, they still do not support real time stream processing, which are one of the most enthralling features in Big Data systems.

Compatible technologies: Polybase [18] is the Microsoft technology enabling queries that target both SQL Server database and Hadoop HDFS storage [6]. The Polybase team recommends the use of a Y Architecture for decisional use cases. In the same context, Oracle Big Data technology [2] uses the same concept, combining both Oracle Exadata with Hadoop HDFS, but they focus more in their approach on the sequential architecture [15].

3.5 Big-Analytics

In this architecture, Big Data technologies are integrated in the analytics phase based on new compute patterns like map-reduce. Usually, these patterns rely on Big Data concepts like distribution and data locality, but these are not always applied in this case because of the non-distribution of the Data-Warehouse. Nevertheless, even without a real distribution of the data, these technologies support a massively parallel processing phase, that improves in a spectacular way the performance of machine learning algorithms. An extraction process splits the input data between a set of workers, triggering a parallel and distributed complex processing.

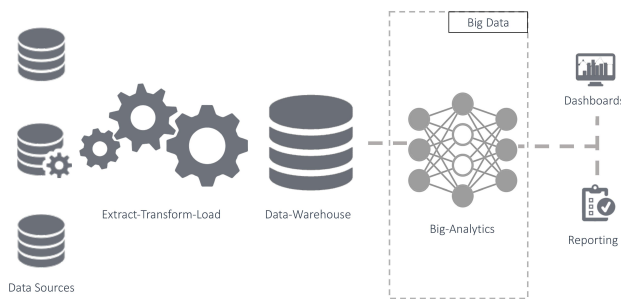


Figure 6. Big-Analytics Architecture

Benefits: The Big-Analytics architecture is characterized by its distributed complex analytics, scalability and graph analytics support. Its biggest benefit is to improve noticeably the performance of Machine Learning algorithms, reducing thus the execution time in a significant way (it can drop from hours to several minutes).

Limits: With Big-Analytics, data locality is not always inevitably supported because of the separation between the Data-Warehouse and the processing system. In addition, these systems are still poor in terms of the number of supported algorithms, because they require a rewriting of machine learning algorithms using the distributed paradigms which is a hard and complex process, and their support from the targeted Big Data processing technologies. For now, we can find a varied choice of implemented algorithms in popular solutions like Spark MLlib or Flink ML, but they are still less varied and numerous than their centralized alternatives.

Compatible technologies: Both Spark [11, 19] and Flink [1] are designed to process data over distributed environments. In fact, they implement a set of complex machine learning and graph computation algorithms characterized by high memory and CPU usage.

3.6 Big-Architecture

Big-Architecture rallies a set of Big Data architectures for decisional projects based only on Big Data Technologies,

patterns and tools. This architecture is seen as a revolution in the decisional world thanks to its modern design, overall scalability and new features, especially real time stream processing.

There are two well known Big Data architectures for decisional projects:

Lambda Architecture. A generic architecture designed to store and process a massive amount of data while taking into account the volume, the velocity and the low latency constraints. This architecture solves complex problems that involve real-time stream processing as well as batch jobs. It aims to provide complete and dynamic views and satisfy the needs for a scalable, robust and fault-tolerant system. It consists mainly of three layers: the first layer is called the Speed Layer, and aims to produce real-time views only on recent streaming data. The data is processed on the fly, without any access to the historic store and only resulting views are stored. The second layer is the Batch Layer, and its main goal is to perform on-demand historic analysis. Data is first stored in a distributed batch storage, as it is, without or with very little loss. Once stored, any analytical request can be performed, and its result will be stored in batch views. Speed Views and Batch Views, resulting from the speed and batch processes, are managed by the Serving Layer, which is responsible of their creation, their maintenance and their eventual fusion and integration. This layer is also the only communication endpoint provided to the decision makers, that can perform batch low-latency analysis along with real-time visualization.

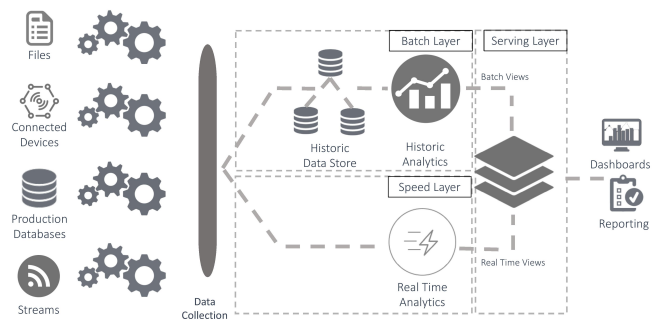


Figure 7. Lambda Architecture

Kappa Architecture. The Kappa architecture is designed in order to simplify the Lambda architecture by combining speed layer and batch layer into one layer known as real-time layer. In fact, this layer is not only used for real-time stream processing, but also to fulfill historic needs, based on both recent and old data, such as real-time machine learning. And since there is no batch processing layer, only one set of

code needs to be maintained, which can decrease the complexity of the system, the biggest hindrance in the lambda architecture.

Because the final user is no longer connected directly to the historic store, he cannot trigger any on-demand processing and all historic analysis must go through the streaming factory.

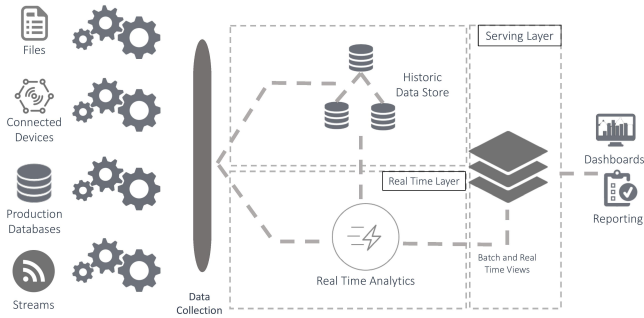


Figure 8. Kappa Architecture

Other Big Architectures. Lambda and Kappa architectures can be the most famous Big Data architectures, but they are not the only ones. In fact, some other less known greek-lettered-architectures exist, such as *iota* [8] which is a lambda-inspired architecture specifically designed for internet of things applications. Other than that, a lot of companies choose to combine Kappa and Lambda architectures in order to benefit from both architectures advantages.

Benefits: The Big Data architectures for decisional projects are as advanced as they are today thanks to their support for real-time Big Data analytics, their overall distribution and scalability and their high availability. In addition, they support schema-less design for historical storage and analytics. On the other hand, these architectures enable us to consider and handle data differently, depending on their types and characteristics. In fact, data can be classified according to its access date, and access and change frequency. So, distinguishing a multi-temperature data management solution involves having **hot**, **warm** and **cold** data, the first being the most recent and frequently accessed data, and the last the oldest and rarely accessed ones. This data classification is very helpful to optimize data access and change operations particularly for Big Data architectures. For example, we can choose the type of storage depending on the type of the data it contains, putting the hot data on stores privileging access speed, while the cold data can be put on stores supporting enormous volumes, but not necessarily fast. In the Lambda architecture, the hot data is immediately processed by the stream layer, while the cold data is archived in the Data-Lake and processed on demand for Analysis, Historical Lookup or Auditing.

Limits: In an existing and implemented decisional project, this architecture is not easy to integrate because it represents a major architecture revolution. Moreover, to implement this architecture, it requires a broad expertise in Big Data systems and technologies, which are known to be less documented and less regulated than other technologies, in addition to the lack of any Big Data methodology for decisional projects. On the other hand, using Big Data technologies can be too complex and extreme for some simple and straightforward use cases where plain old BI solutions can do the job. If real-time is not needed, if KPIs are clear, fixed, and well-defined from the beginning, there is no need to go for a haphazard Big Data decisional architecture with less support and documentation.

Compatible technologies: Some of the well known OLAP decisional technologies that respect the Lambda architecture is Druid[20]. Indeed, it is an open source, end-to-end solution meant to offer flexible, low-latency analytic queries on real-time data. The solution combines the low latency guarantees of stream processors and the correctness and flexibility guarantees of batch processors. It also introduces a serving layer specifically designed for interactive analytics. Another technology, called Samoa [3] implements the Kappa Architecture, by introducing machine learning models in the streaming context, combining both batch and streaming jobs into the same layer.

4 How to make a choice?

For most companies and researchers that have to implement a decisional Big Data architecture, choosing the right setting is the most difficult task, along with the integration of the different components. In fact, the lack of a clear reference documentation presenting the pros and cons of each choice leads each architect to implement his own version of the architecture, which provides as a result draft solutions, mostly inadequate with their needs, and so difficult to implement and test that they end up abandoning them on the long run.

In order to simplify making the decisional architecture choice, we represent a capability matrix in Tab.2 that summarizes different supported features for the cited architectures and the differences between them.

The set of criteria we are considering are the following:

- **Real-time Stream Processing:** It is the ability to collect, store and analyze streaming data in stream real-time with a low latency.
- **Data-driven Historic Analytics:** This is the main decisional feature of the data lake allowing for data collection and storage without any losses. It aims to make all decisional requests and KPIs possible to implement on-demand.
- **High Volume Storage:** Classical Data-Warehouses are supposed to deal with a Big Data volume, but being

centralized, they can reach their limits very quickly if the data keeps on growing constantly.

- **High Volume Extraction:** Some systems are confronted to a set of various data sources with voluminous generated data. We talk about high extraction volume when we have more than 1TB to extract per day from one or many sources.
- **Complex Analysis:** This characteristic represents the capacity of the system to make massive parallel computing. Complex analysis can be performed on distributed or non distributed data storage, and supports new paradigms and patterns to boost the performance of systems using complex analysis, which are very demanding in terms of system resources (RAM and CPU), like for example Machine Learning algorithms or Graph computations.

Based on this capability matrix, we represent a decision tree (Fig.9) to help architects decide which solution to adopt depending on their needs. The first thing to check in this decision tree is whether if real time processing is needed. In this case, only Big-Architecture can respond to this kind of feature. If targeted solutions only deal with classical batch processing, we must check the type of request needs, whether they are KPI-driven (defined at the elaboration of the architecture) or data-driven (depending on the available data), because only Data-Lake $\cup$ Data-Warehouse architecture among the remaining architectures supports data-driven historic analytics. Then, the stored data volume will impose the choice of Big-Warehouse architecture if it exceeds 100 TB, as it is the only solution among the remaining ones that supports distributed storage. Finally, the choice between Big-ETL, Big-Analytics and Big-Architecture will be impacted by the extracted data volume and analysis complexity. In fact, if the data to be extracted is not voluminous, but the analysis is complex and needs a lot of resources, Big-Analytics can be the best alternative. But if the extracted data is big, and the processing quite simple without a need of a lot of resources, Big-ETL can be sufficient. When the selected architecture is a combination of Big-ETL and Big-Analytics, we propose to replace them with Big-Architecture.

Lastly, let's point out that the Big-Architecture is the most powerful architecture thanks to its completeness and its capacity to ensure all the matrix features. So, it can replace any decisional architecture and make it more efficient. But due to its complexity, and for the sake of reuse, it is sometimes better to use other architectures. When an existing BI architecture is already deployed, it is easier to integrate the Big-ETL, Big-Analytics and/or Data-Lake $\cup$ Data-Warehouse architecture where and when needed.

5 Conclusion

In our paper, we summarize the various decisional architectures combining Big Data and Business Intelligence concepts

and tools depending on the business needs. We also provide a decision tree in order for architects to choose the most suitable alternative for their systems.

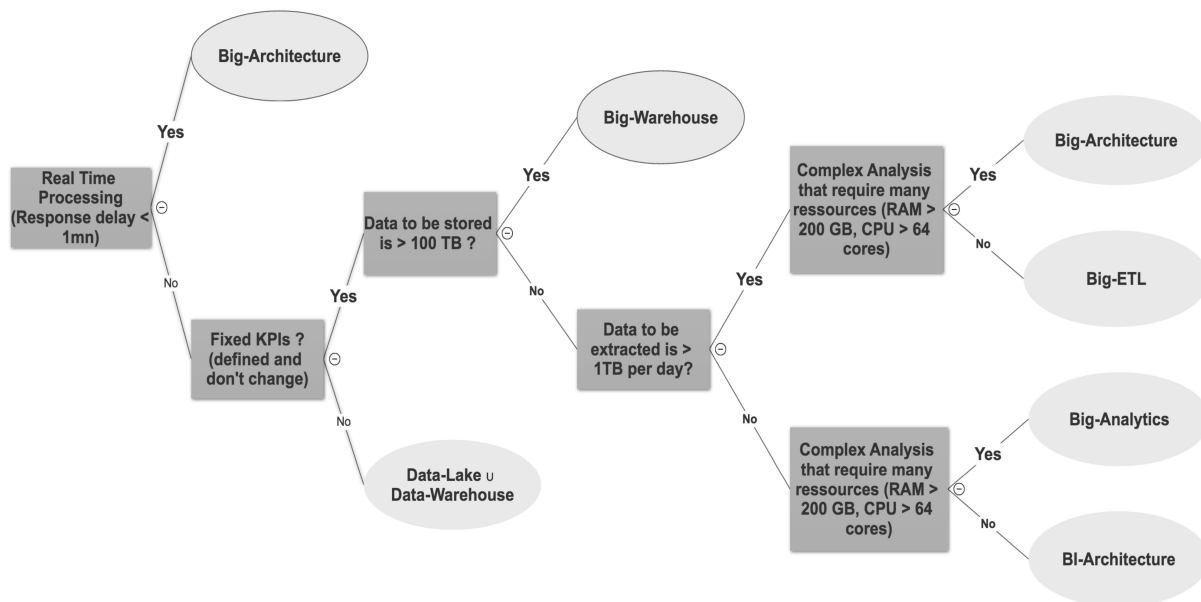
We notice that pure decisional Big Data architectures can cover all the features of traditional BI architectures, adding into consideration the storage distribution and the processing complexity. However, in the case where decisional needs can be covered by a BI architecture, choosing the latter remains more coherent and easier to set up. The fact that pure Big Data architectures choose a zero-loss policy, that they do not have a well-defined and clear data structure (like data-warehouses do), and that they have governance and exploration issues due to the variety and the low density of the data, makes these systems very difficult to implement and control, and makes value extraction a very hard task to perform. These difficulties call for the definition of design and architectural patterns for Big Data systems, along with the best practices to follow, and the anti-patterns to avoid, when designing, developing and maintaining these types of systems.

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Table 2. Comparative Capability Matrix

	Real Time Stream Processing	Data-driven Historic Analytics	High Volume Storage	High Volume Extraction	Complex Analysis
BI					
Big-ETL				X	
Big-Analytics					X
Big-Warehouse			X	X	X
D-L $\cup$ D-W		X	X	X	X
Big-Arch	X	X	X	X	X

**Figure 9.** Decision Tree for Decisional Architectures

- [data.html](#)
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